

Do you need help with interpreting the plots? See [here](#).

PX4 VTOL Standard

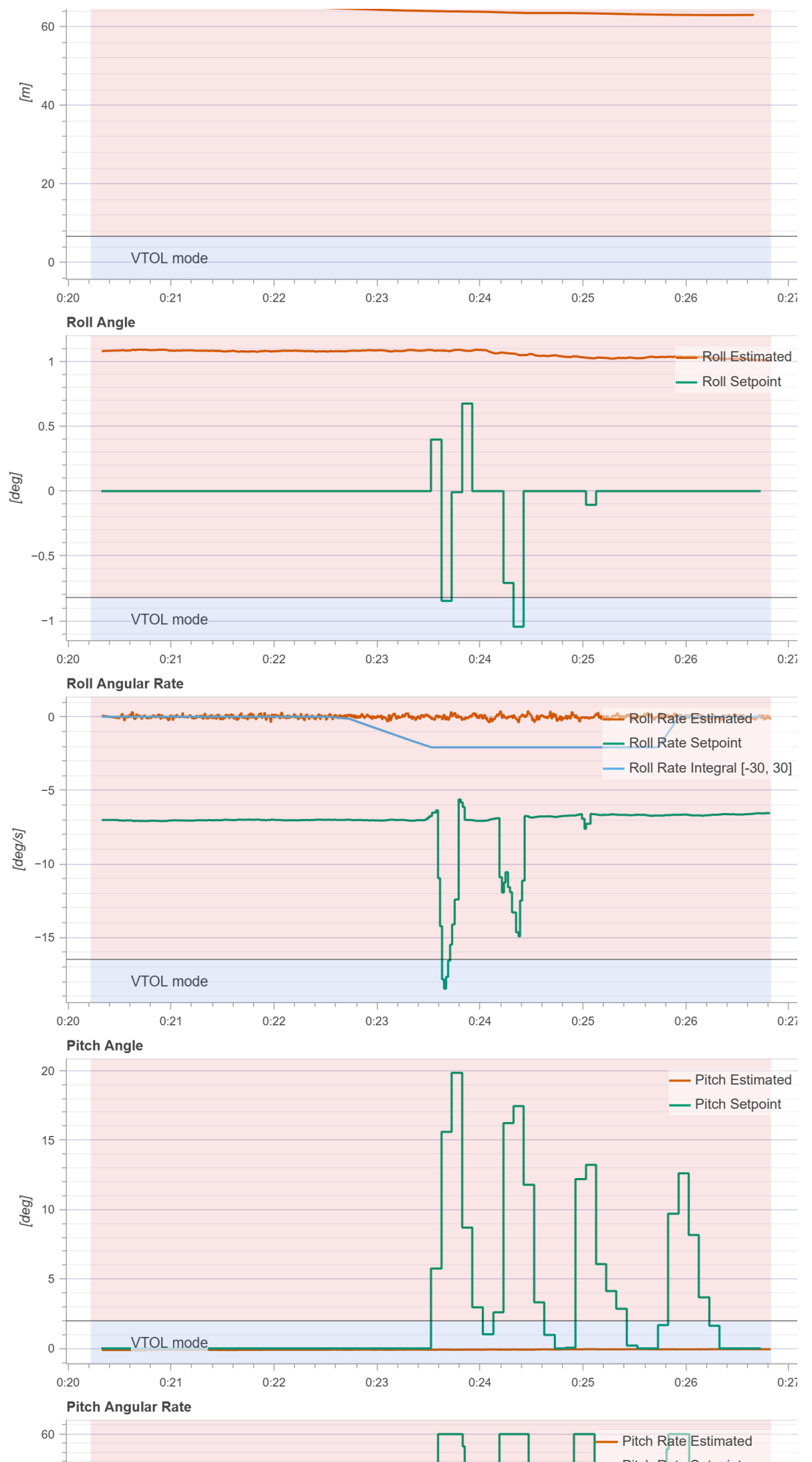
[Open 3D View](#)[Open PID Analysis](#)

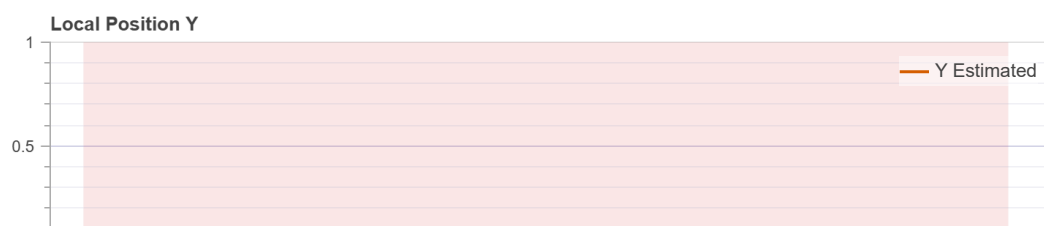
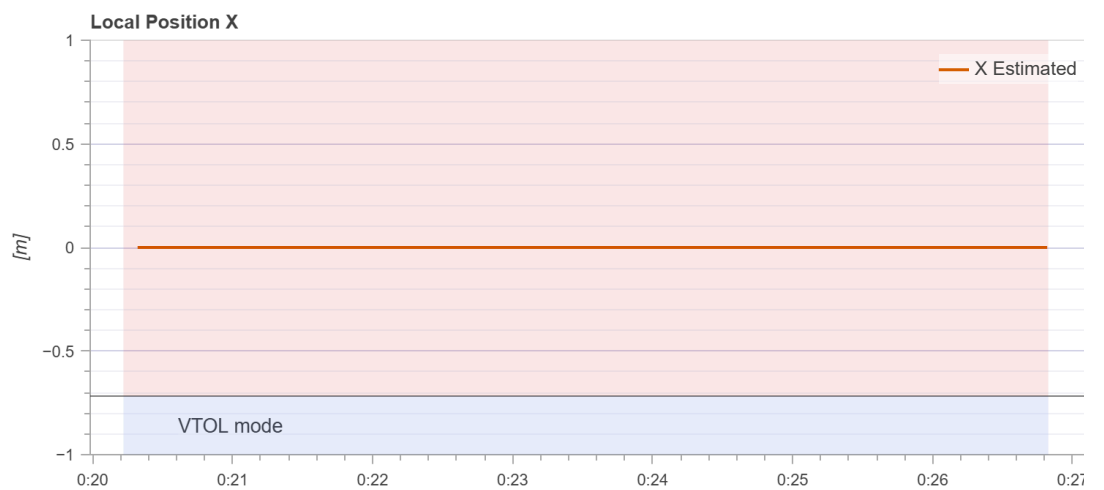
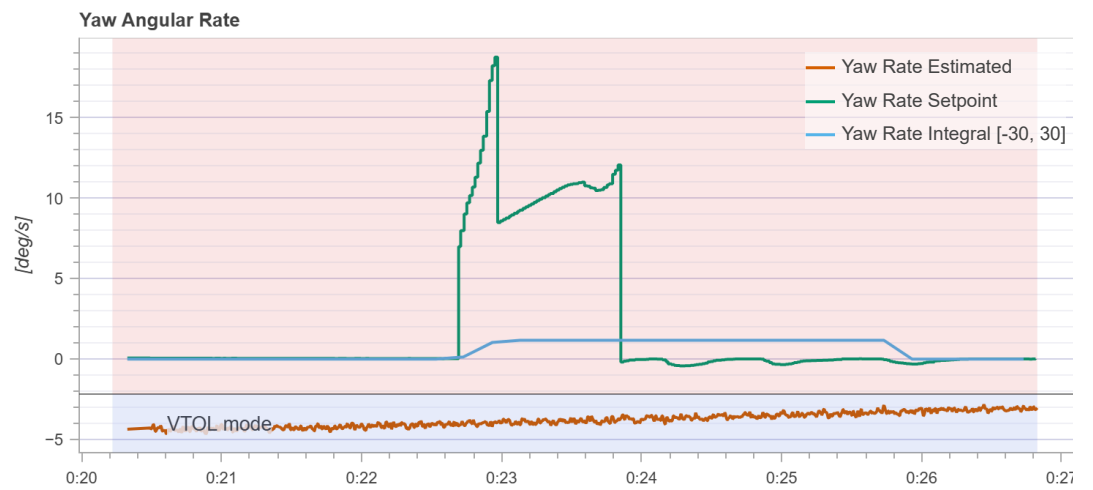
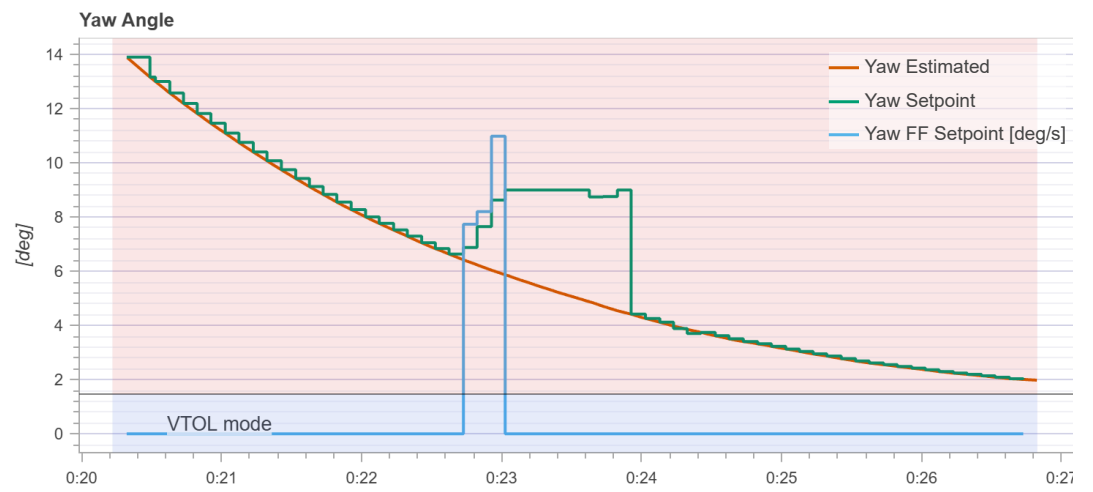
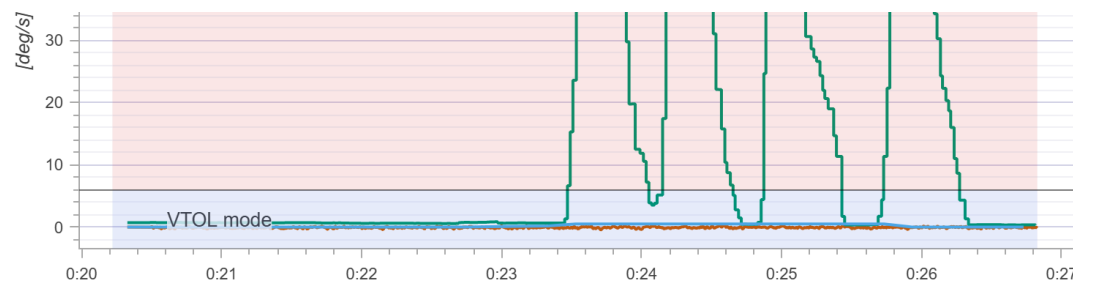
Airframe:	13014		
Hardware:	CUBEPILOT_CUBEORANGE		
Software	8583f1da		
Version:	branch: v1.11.2_w_rc_sysid		
OS		Max	
Version:	NuttX, v8.2.0	Altitude	1
Estimator:	EKF2	Difference:	m
Logging			
Start ?:	21-04-2021 12:00		
Logging			
Duration:	0:00:06		
Dropouts:	1 (0.03 s)		
Vehicle			
Life	6 hours 7 minutes 15 seconds	Max Tilt	1.1
Flight		Angle:	deg
Time:			
Vehicle			
UUID:	000600000000383638393239510d0035002d		

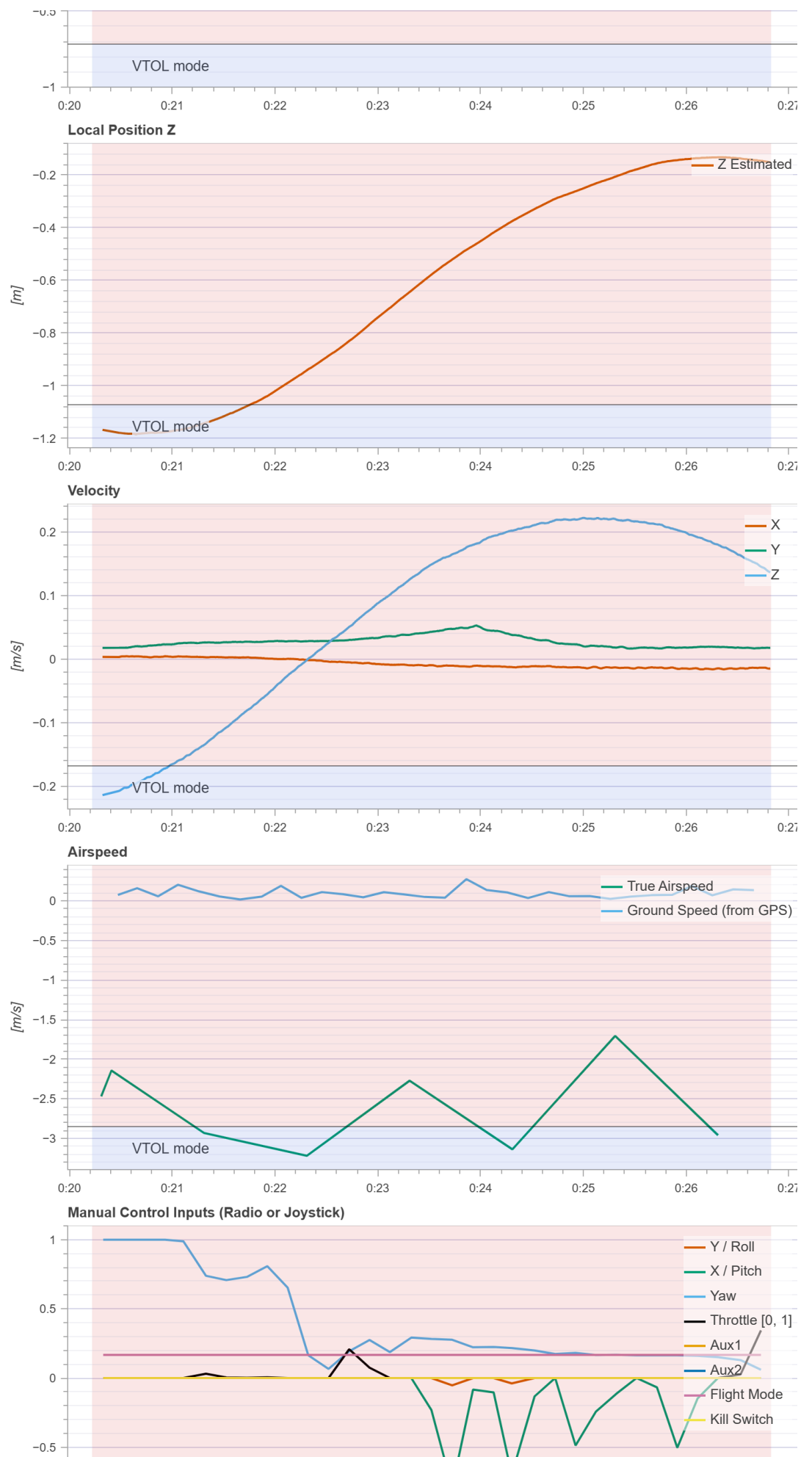
Select Some Options

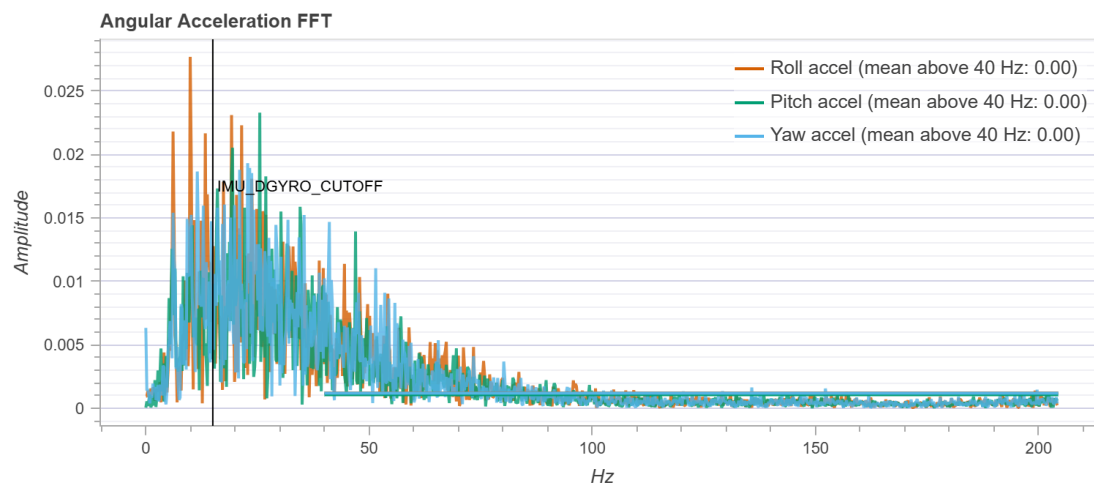
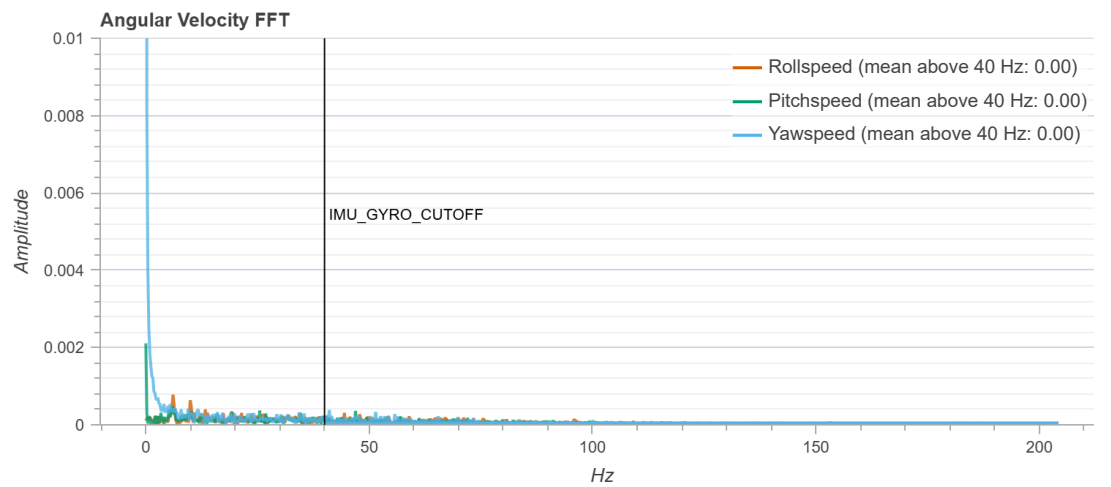
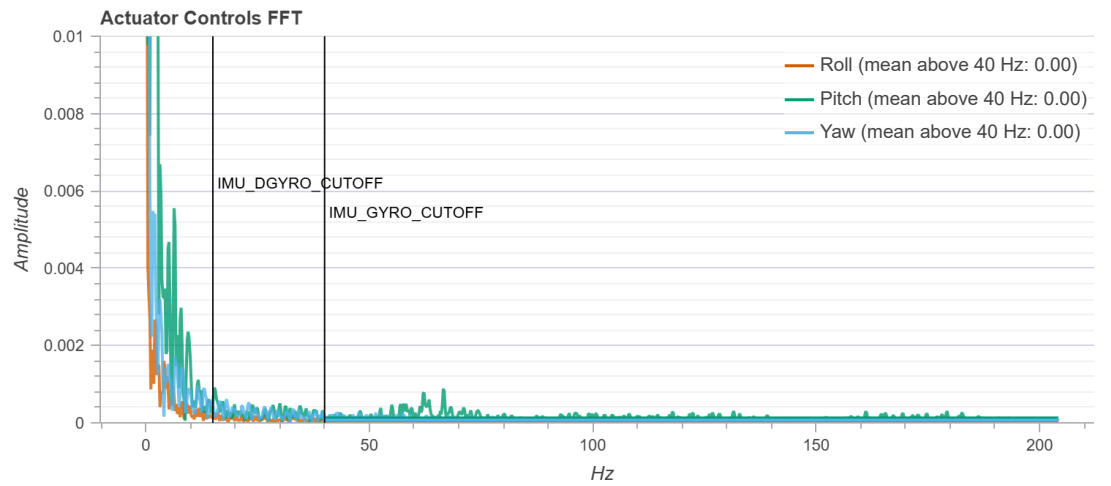
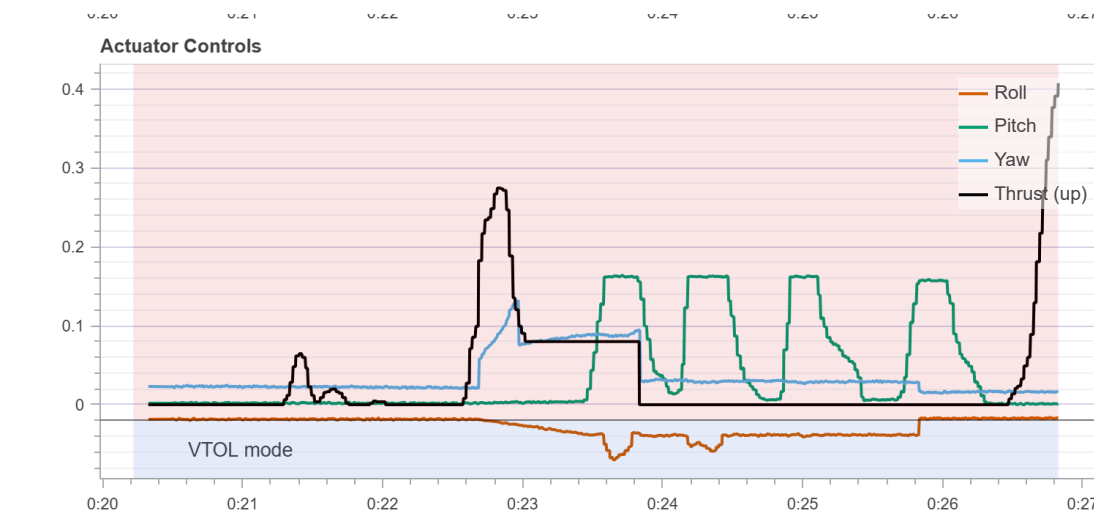


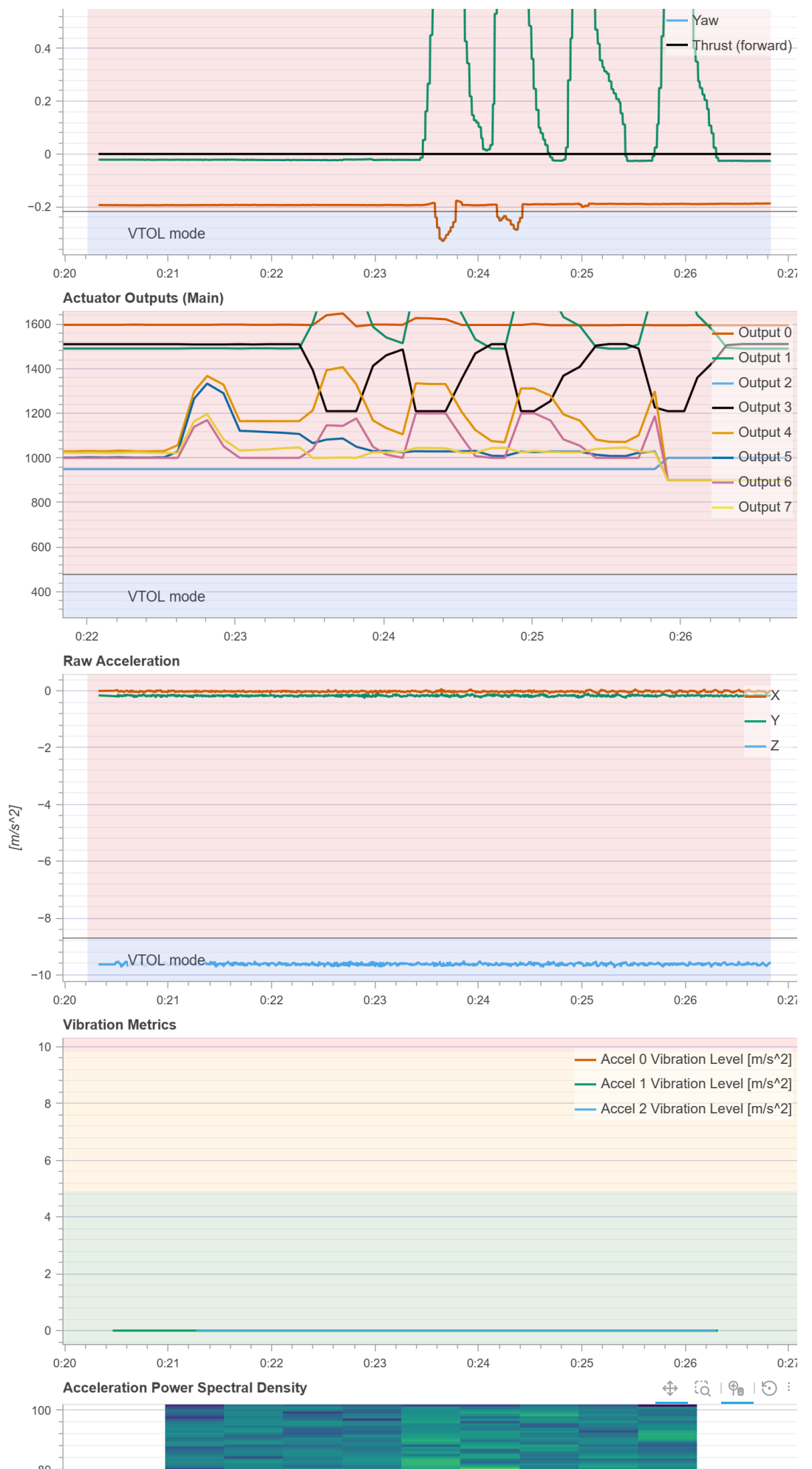
Altitude Estimate

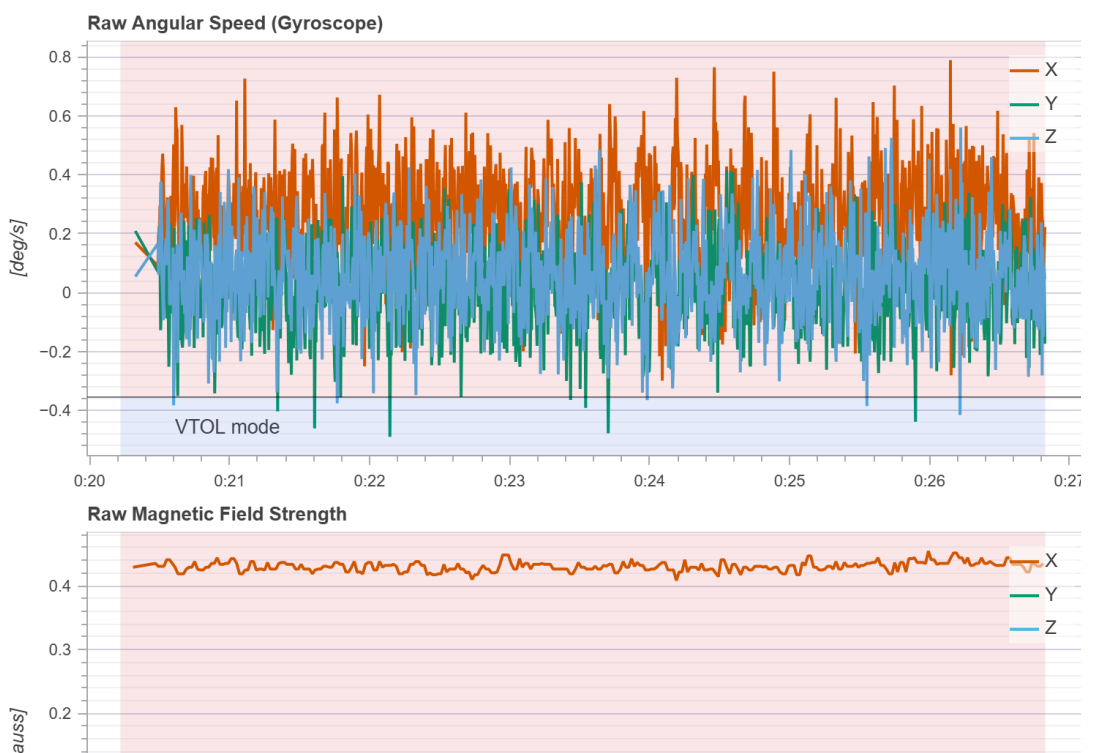
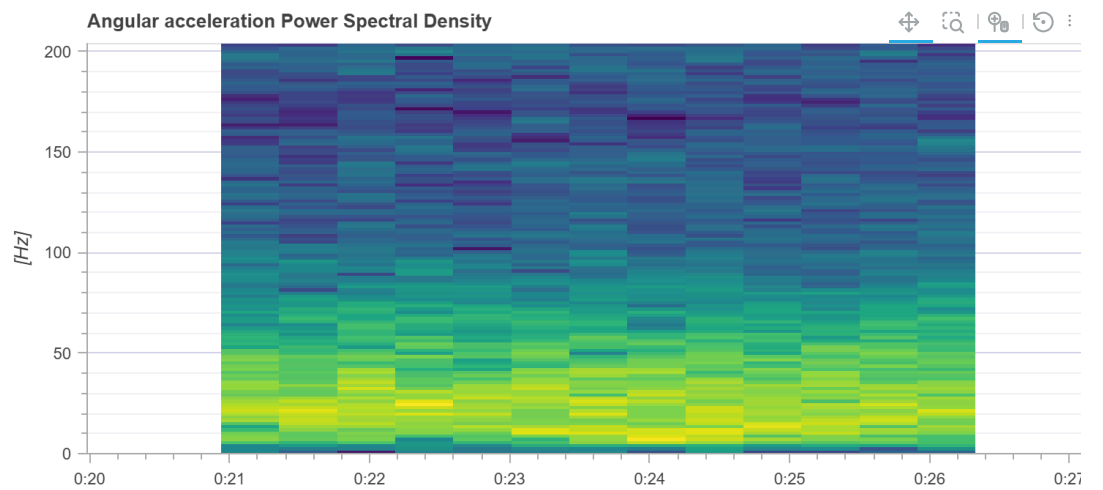
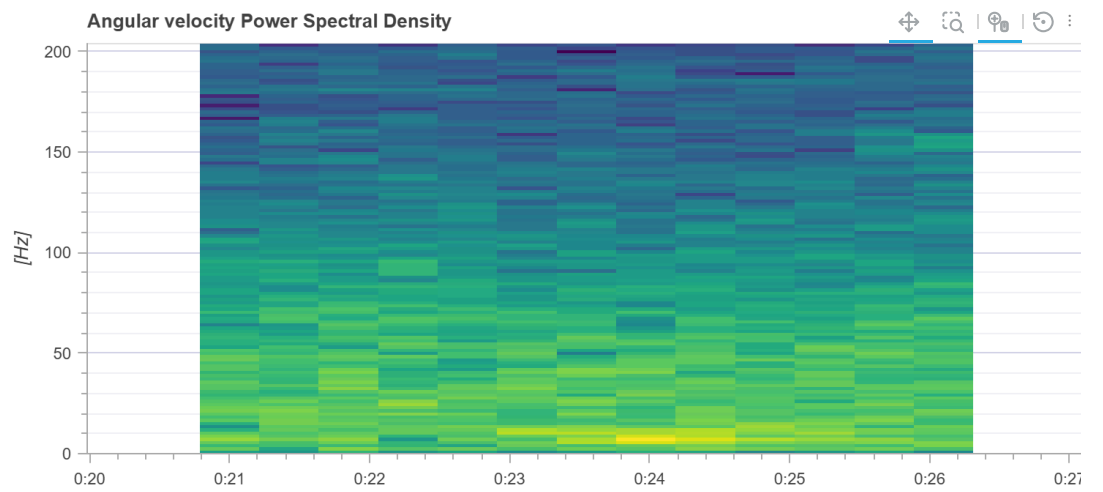
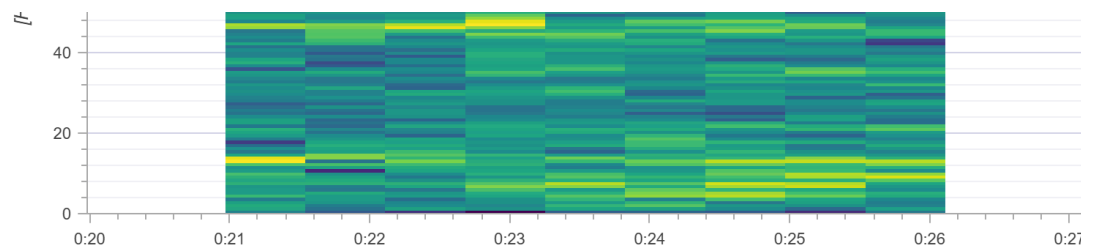




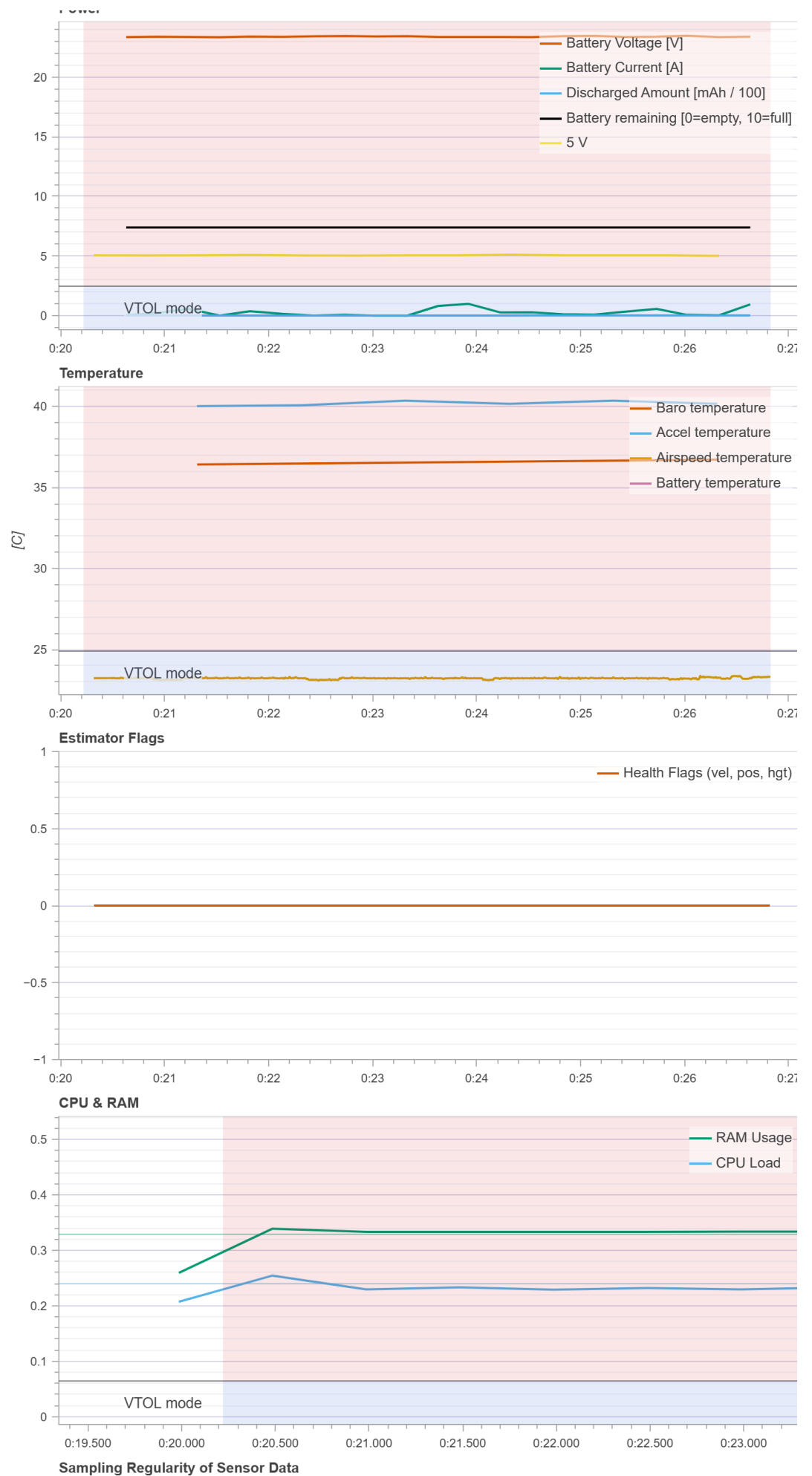


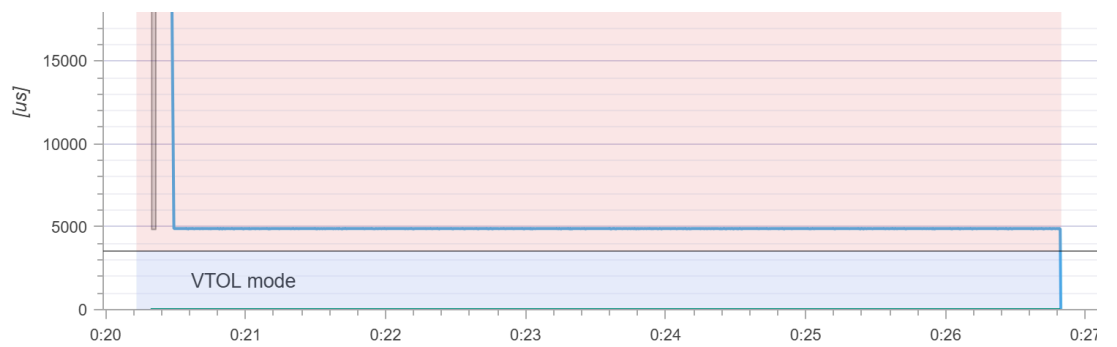












Non-default Parameters (except RC and sensor calibration)

#	Name	Value	Default	Min	Max	Description
0	ASPD_BETA_GATE	1				(unknown)
1	ASPD_BETA_NOISE	0.30000001192092...				(unknown)
2	ASPD_DO_CHECKS	0	7	0	31	Enable checks on airspeed sensors
3	ASPD_FS_INNOV	1	5	0.5	10.0	Airspeed failure innovation threshold
4	ASPD_FS_INTEG	-1	10	0.0	50.0	Airspeed failure innovation integral th
5	ASPD_FS_T1	3				(unknown)
6	ASPD_FS_T2	100				(unknown)
7	ASPD_SCALE	1				(unknown)
8	ASPD_SCALE_EST	0				(unknown)
9	ASPD_SC_P_NOISE	0.00009999999747...				(unknown)

Logged Messages

#	Time	Level	Message
0	0:00:22	INFO	[commander] Takeoff detected
1	0:00:23	INFO	[commander] Landing detected
2	0:00:25	INFO	[commander] Disarmed by landing

Hide additional Data

Console Output

```
sercon: Registering CDC/ACM serial driver
sercon: Successfully registered the CDC/ACM serial driver
HW arch: CUBEPILOT_CUBEORANGE
FW git-hash: 8583f1da30b63154d6ba0bc187d86135dfe33cf9
FW version: 1.11.2 0 (17498624)
FW git-branch: v1.11.2_w_rc_sysid
OS: NuttX
OS version: Release 8.2.0 (134349055)
OS git-hash: ec20f2e6c5cc35b2b9bbe942dea55eabb81297b6
Build datetime: Mar 22 2021 17:59:08
Build uri: localhost
Toolchain: GNU GCC, 9.3.1 20200408 (release)
PX4GUID: 000600000000383638393239510d0035002d
MCU: STM32H7[4|5]xxx, rev. V
INFO [param] selected parameter default file /fs/mtd_params
INFO [tune control] Publishing standard tune 1
```

```

ms5611 #0 on SPI bus 1 (devid=0x3d)
icm20649 #0 on SPI bus 1 (devid=0x25)
ms5611 #1 on SPI bus 4 (devid=0x3d)
icm20602 #0 on SPI bus 4 (devid=0x38)
icm20948 #0 on SPI bus 4 (devid=0x28)
ak09916 #0 on I2C bus 1 (external, equal to '-b 2')
WARN [SPI_I2C] Already running on bus 1
ms4525_airspeed #0 on I2C bus 2 (external, equal to '-b 1')
Board extras: /etc/init.d/rc.board_mavlink
INFO [mavlink] mode: Config, data rate: 800000 B/s on /dev/t
Starting Main GPS on /dev/ttyS2
Starting MAVLink on /dev/ttyS0
INFO [mavlink] mode: Normal, data rate: 1200 B/s on /dev/tty
Starting MAVLink on /dev/ttyS1
INFO [mavlink] mode: Onboard, data rate: 2880 B/s on /dev/tt
INFO [px4io] default PWM output device
INFO [init] Mixer: /etc/mixers/babyshark.main.mix on /dev/pw
INFO [init] Mixer: /etc/mixers/pass.aux.mix on /dev/pwm_outp
INFO [fw_att_control] 1405213 Sysid initialized
INFO [logger] logger started (mode=all)

NuttShell (NSH)
nsh> [KINFO [ecl/EKF] reset position to last known position
INFO [ecl/EKF] reset velocity to zero
INFO [ecl/EKF] 9270059: EKF aligned, (baro hgt, IMU buf: 18,
WARN [PreFlightCheck] Preflight: GPS Horizontal Pos Error to
INFO [logger] Start file log (type: full)
INFO [logger] [logger] /fs/microsd/log/2021-04-21/06_30_58.u
INFO [logger] Opened full log file: /fs/microsd/log/2021-04-

```

Processes

Pre Flight:

PID	COMMAND	CPU(ms)	CPU(%)	USED/STACK	PR
0	Idle Task	16387	76.700	200/ 512	0
1	hpwork	0	0.000	344/ 1260	245
2	lpwork	0	0.000	344/ 1612	50
4	wq:manager	0	0.000	384/ 1252	255
498	log_writer_file	10	0.000	608/ 1164	60
17	dataman	27	0.000	768/ 1204	90
19	wq:lp_default	5	0.000	836/ 1700	205
23	wq:I2C1	56	0.200	892/ 1468	240
28	wq:hp_default	190	1.000	1044/ 1900	240
140	wq:SPI1	401	2.000	1616/ 2332	253
143	wq:SPI4	730	3.700	1616/ 2332	250
197	wq:I2C2	80	0.400	836/ 1468	245
200	wq:nav_and_controllers	886	4.800	5288/ 7196	245
201	wq:rate_ctrl	315	1.600	1108/ 1660	255
208	commander	120	0.400	1780/ 3212	140
210	commander_low_prio	0	0.000	716/ 2996	50
270	gps	17	0.100	968/ 1676	205
307	mavlink_if1	111	0.500	1856/ 2484	100
308	mavlink_rcv_if1	36	0.200	2548/ 4068	175
317	mavlink_if2	238	1.200	1872/ 2484	100
318	mavlink_rcv_if2	31	0.200	2544/ 4068	175
344	px4io	426	2.100	1036/ 1484	237
461	wq:attitude_ctrl	108	0.900	1212/ 1668	242

```
Processes: 25 total, 2 running, 23 sleeping, max FDs: 15
CPU usage: 21.40% tasks, 1.90% sched, 76.70% idle
DMA Memory: 5120 total, 1536 used 1536 peak
Uptime: 21.326s total, 16.387s idle
```

Performance Counters

Pre Flight:

```
vehicle_imu: gyro data gap: 1 events
vehicle_imu: gyro update interval: 7612 events, 2443.69 avg,
vehicle_imu: accel data gap: 1 events
vehicle_imu: accel update interval: 7612 events, 2443.69 avg,
vehicle_imu: gyro data gap: 1 events
vehicle_imu: gyro update interval: 7442 events, 2499.66 avg,
vehicle_imu: accel data gap: 1 events
vehicle_imu: accel update interval: 7442 events, 2499.66 avg,
vehicle_imu: gyro data gap: 1 events
vehicle_imu: gyro update interval: 7614 events, 2443.68 avg,
vehicle_imu: accel data gap: 1 events
vehicle_imu: accel update interval: 7614 events, 2443.68 avg,
logger_sd_fsync_mission: 0 events, 0us elapsed, 0.00us avg, r
logger_sd_write_mission: 0 events, 0us elapsed, 0.00us avg, r
logger_sd_fsync: 0 events, 0us elapsed, 0.00us avg, min 0us r
logger_sd_write: 0 events, 0us elapsed, 0.00us avg, min 0us r
navigator: 229 events, 2145us elapsed, 9.37us avg, min 5us ma
land_detector: cycle: 1249 events, 11118us elapsed, 8.90us av
fw_pos_control_l1: cycle: 546 events, 5813us elapsed, 10.65us
fw_att_control: cycle: 2226 events, 55683us elapsed, 25.01us
mc_hover_thrust_estimator: cycle time: 0 events, 0us elapsed,
mc_pos_control: cycle time: 546 events, 3605us elapsed, 6.60u
mc_att_control: cycle: 2226 events, 35798us elapsed, 16.08us
mc_rate_control: cycle: 7641 events, 56744us elapsed, 7.43us
vtol_att_control: cycle: 7640 events, 100562us elapsed, 13.16
airspeed_selector: elapsed: 184 events, 2887us elapsed, 15.69
ekf2: update: 3780 events, 260483us elapsed, 68.91us avg, min
pwm_out: interval: 864 events, 21934.12 avg, min 190us max 33
pwm_out: cycle: 864 events, 11132us elapsed, 12.88us avg, min
control latency: 0 events, 0us elapsed, 0.00us avg, min 0us r
io control latency: 7423 events, 4348538us elapsed, 585.82us
io write: 0 events, 0us elapsed, 0.00us avg, min 0us max 0us
io update: 7426 events, 5558917us elapsed, 748.57us avg, min
io_txns: 21806 events, 5386948us elapsed, 247.04us avg, min
mavlink: send_start tx buffer full: 25 events
mavlink: send_bytes error: 0 events
mavlink: tx run interval: 3773 events, 5003.44 avg, min 4209
mavlink: tx run elapsed: 3773 events, 248134us elapsed, 65.77
mavlink: send_start tx buffer full: 10 events
mavlink: send_bytes error: 0 events
mavlink: tx run interval: 1887 events, 9999.47 avg, min 9265
mavlink: tx run elapsed: 1887 events, 111214us elapsed, 58.94
vehicle_air_data: cycle: 1418 events, 16711us elapsed, 11.78
mavlink: send_start tx buffer full: 0 events
mavlink: send_bytes error: 0 events
mavlink: tx run interval: 0 events, 0.00 avg, min 0us max 0us
mavlink: tx run elapsed: 0 events, 0us elapsed, 0.00us avg, r
battery_status: 1910 events, 30988us elapsed, 16.22us avg, m
sensors: 3831 events, 129689us elapsed, 33.85us avg, min 11us
```

```
ak09916: bad transfer: 0 events
ak09916: bad register: 0 events
ak09916: transfer: 950 events, 1075418us elapsed, 1132.02us avg, min 0us max 1132.02us
icm20948_ak09916: magnetic sensor overflow: 0 events
icm20948_ak09916: bad transfer: 1 events
icm20948: FIFO reset: 1 events
icm20948: FIFO overflow: 0 events
icm20948: FIFO empty: 0 events
icm20948: bad transfer: 0 events
icm20948: bad register: 0 events
icm20602: FIFO reset: 1 events
icm20602: FIFO overflow: 0 events
icm20602: FIFO empty: 0 events
icm20602: bad transfer: 0 events
icm20602: bad register: 0 events
ms5611: com_err: 0 events
ms5611: measure: 1912 events, 6570us elapsed, 3.44us avg, min 0us max 6570us
ms5611: read: 1911 events, 22237us elapsed, 11.64us avg, min 0us max 22237us
icm20649: DRDY missed: 0 events
icm20649: FIFO reset: 1 events
icm20649: FIFO overflow: 0 events
icm20649: FIFO empty: 0 events
icm20649: bad transfer: 0 events
icm20649: bad register: 0 events
ms5611: com_err: 0 events
ms5611: measure: 1916 events, 6493us elapsed, 3.39us avg, min 0us max 6493us
ms5611: read: 1915 events, 24784us elapsed, 12.94us avg, min 0us max 24784us
adc: sample: 13489 events, 108423us elapsed, 8.04us avg, min 0us max 108423us
rc_update: 861 events, 11489us elapsed, 13.34us avg, min 10us max 11489us
load_mon: cycle: 39 events, 2164us elapsed, 55.49us avg, min 0us max 2164us
dataman: write: 0 events, 0us elapsed, 0.00us avg, min 0us max 0us
dataman: read: 5 events, 21033us elapsed, 4206.60us avg, min 0us max 21033us
dma_alloc: 4 events
param_set: 240 events, 6217us elapsed, 25.90us avg, min 1us max 6217us
param_get: 9258 events, 15015us elapsed, 1.62us avg, min 1us max 15015us
param_find: 6594 events, 14882us elapsed, 2.26us avg, min 1us max 14882us
param_export: 0 events, 0us elapsed, 0.00us avg, min 0us max 0us
```